

Rofu Thailand rooftop solar and storage

Feasibility results on the
measured roof: five configurations
on the PEA time-of-use tariff, sized
by optimisation and tested on a 20
year pro forma

Prepared by Allotrope Partners
September 2026



Summary: roof F pays alone; all roofs earn four times as much

16,319 m2

measured roof on seven buildings; roof F (3,329 m2, replaced 1 to 2 years ago) carries 433 kWp, all roofs 2,122 kWp

433 / 2,122 kWp

the two roof groups; storage is immaterial on roof F (50 kWh) and selected on all roofs (303 kW / 634 kWh)

77k-357k USD

year-1 bill savings, roof F alone to all roofs with storage, against a business-as-usual bill of 843,443 USD

0.39-1.61 M USD

equity NPV at 11 percent over 20 years; equity IRR 60.3% to 75.5% on 70 percent debt

What the analysis found

- Roof F alone is a sound first project: 216,400 USD of capex, 77,247 USD of year-1 savings, equity IRR 75.5%, and almost no curtailment (1.4%) because the array is small against a 1.15 MW daytime load.
- The other six roofs hold four fifths of the value: all roofs with storage lifts NPV from 0.39 to 1.61 M USD, at the cost of a condition review those roofs have not had.
- Storage only earns at scale. On roof F the optimiser adds a token 50 kWh; on all roofs it adds 634 kWh, recovering curtailed energy (13.2% without storage, 9.0% with) and clipping the monthly peak.
- Every return still scales off the 500 USD/kWp quotation, below published Thai benchmarks; the price is worth confirming before anything else.

Recommended path: two phases

- Phase 1, roof F, PV only (case 1): 433 kWp, 216,400 USD, NPV 389,829 USD. No structural question; can proceed to an EPC quotation now.
- Phase 2, the six older roofs with storage (case 4 less case 1): about 1,689 kWp and 634 kWh more, 969,694 USD more capex, 1,222,965 USD more NPV; conditional on the roof condition review.
- Reference: the unconstrained optimum (case 5) remains 2,847 kWp; the measured roof carries three quarters of it.

Question

How much rooftop solar, with or without battery storage, should Rofu Thailand install under its current PEA tariff, and what does the investment return over 20 years?

Method

- Dispatch and sizing: REopt optimisation at 15 minute resolution, one full year of metered load, PVWatts irradiance for Phimai, no grid export.
- Economics: 20 year pro forma, direct ownership by the factory, 70 percent debt, Thai CIT and depreciation.
- Five cases: roof F and all roofs, each without and with storage, plus the unconstrained storage optimum from the memo.

What changed since the 12 September memo

1

The roof is measured

Seven buildings traced on satellite imagery: 16,319 m² in total, of which roof F is 3,329 m². The memo worked from a survey estimate of 12,960 to 16,200 m².

Capacity at the memo's 65 percent and 0.20 kWp/m²: 433 kWp on roof F, 2,122 kWp on all roofs

2

Roof condition is known

Roof F was replaced one to two years ago after unreparable leaks. The other six roofs are older, no replacement is planned, and their suitability has not been assessed.

The cases split into roof F alone and all seven roofs after a condition review

3

Four cases re-run, one carried over

PV only and PV with storage on roof F and on all roofs, all other inputs as in the memo. Case 5 is the memo's unconstrained storage case, unchanged.

Five workbooks, one per case, replace the memo's six

The facility: Rofu Thailand, Phimai

Supply

Utility: Provincial Electricity Authority, Phimai branch

Tariff: Schedule 4.2 Large General Service, time of use, 22-33 kV, rate code 4224

Connection: one grid connection, four transformers, 3,230 kVA installed (500 + 500 + 1,600 + 630)

Location: Phimai district, Nakhon Ratchasima

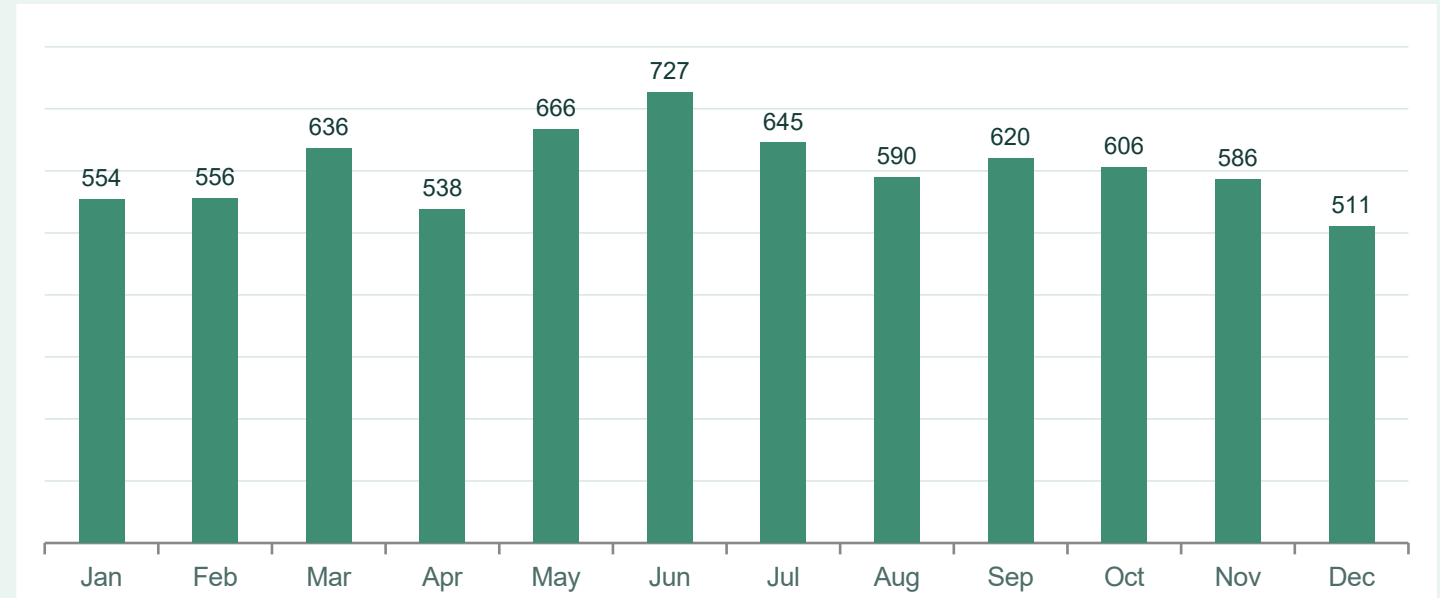
Roof

Measured: seven buildings, 16,319 m² of roof from satellite imagery (next slide)

Roof F: 3,329 m², replaced one to two years ago after unrepairable leaks; the newest roof on site

Roofs A to E and G: older, no replacement planned; condition to be reviewed before a solar decision

Metered electricity use, 2025 (MWh per month)



7.24 GWh

annual consumption

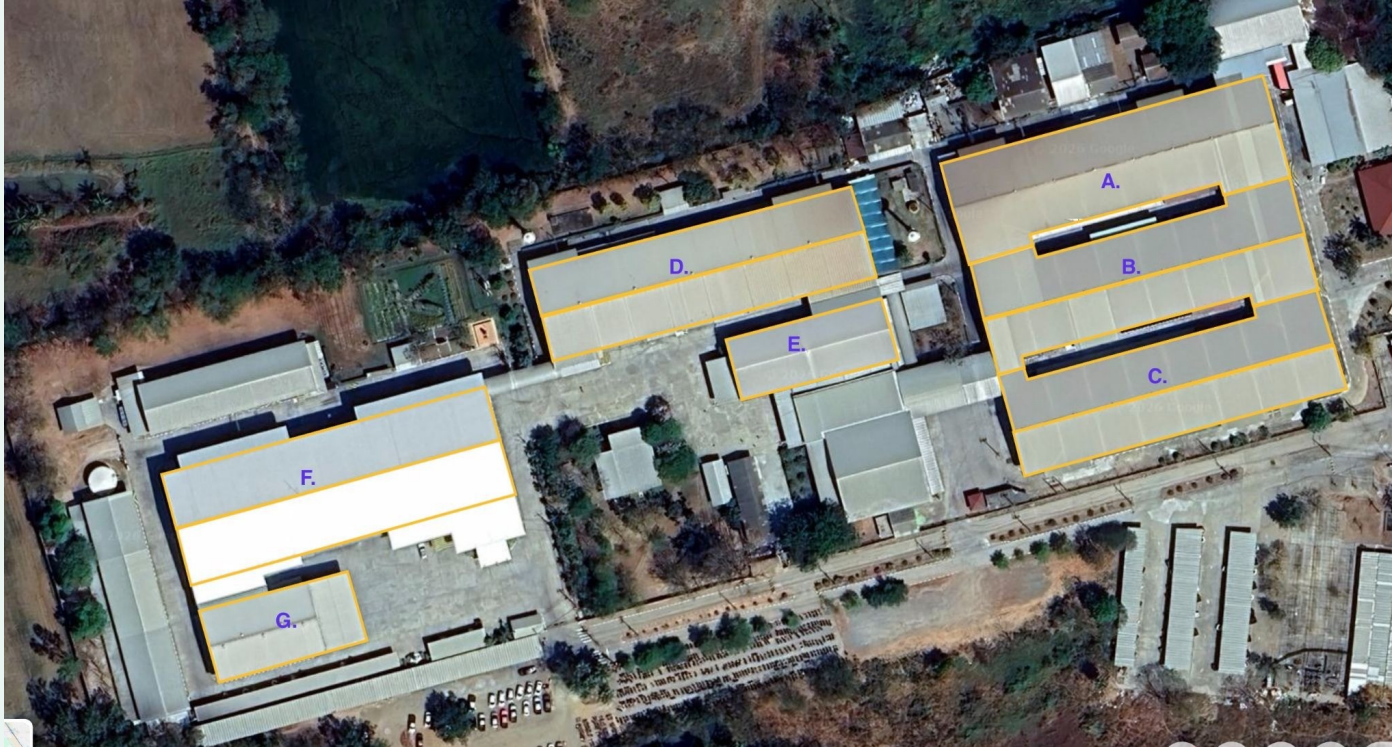
1.21-1.38 MW

monthly on-peak
maximum demand

843,443 USD

business-as-usual bill per
year (THB 27.4 million)

The roof: seven buildings measured from satellite imagery



Satellite view with the seven roof outlines (A to G). Roof F, front of the site, is the replaced roof.

16,319 m2 outlined

x 65% usable

x 0.20 kWp/m2

= 2,122 kWp all
roofs

Roof F alone: $3,329 \text{ m}^2 \times 65\% \times 0.20 = 433 \text{ kWp}$. The memo's survey estimate (five roofs, 12,960 to 16,200 m2 gross, 1,685 to 2,106 kWp) is superseded; the measured total sits at the top of that range.

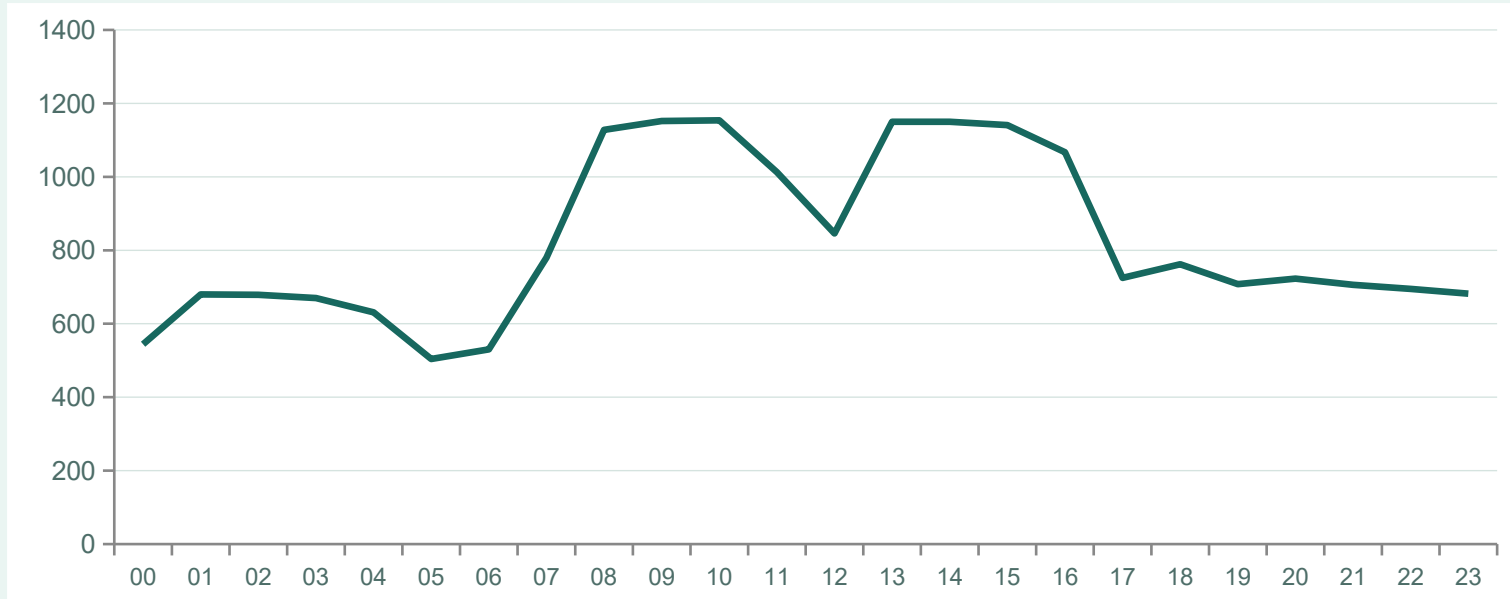
Roof	Area (m2)	kWp	Condition
A	2,818	366	older, to be reviewed
B	2,859	372	older, to be reviewed
C	2,733	355	older, to be reviewed
D	2,720	354	older, to be reviewed
E	885	115	older, to be reviewed
F	3,329	433	new (1 to 2 years)
G	975	127	older, to be reviewed
Total	16,319	2,122	seven buildings

What the roof condition means

- Roof F is the one surface that can carry panels without a structural question: new sheeting, no leak history, about one fifth of the total area.
- Roofs A to E and G hold the other four fifths. Their age and condition have not been assessed; a structural and sheeting review is a precondition, not a formality.
- The cases therefore come in two groups: roof F alone, and all seven roofs once the review clears them.

Daily load against the tariff clock and the solar day

Average load by hour, 2025 (kW)



Weekday tariff

off-peak 2.6037

peak 4.1839 THB/kWh + monthly demand charge

off

Solar output

generation 06:00 to 18:00, peak 11:00 to 14:00

00:00

06:00

09:00

12:00

18:00

22:00

24:00

Weekends and Thai public holidays are off-peak all day.

What this means for the design

- Solar displaces the 4.18 THB peak energy from 09:00 to 18:00 on weekdays, the most valuable kWh on the bill.
- The 08:00 to 16:00 load plateau of about 1.15 MW is above the roof-limited array's average output, so most weekday solar is used on site.
- The demand charge is set by one interval a month; PV alone cannot guarantee a lower peak, storage can.
- Weekends and holidays are off-peak all day and the load is lower: that is where curtailment happens without export.

The PEA tariff: five building blocks, one monthly peak

Peak energy

4.1839 THB/kWh

weekdays 09:00-22:00

Off-peak energy

2.6037 THB/kWh

nights, weekends, holidays

On-peak demand

132.93 THB/kW a month

highest on-peak interval

Ft adjustment

0.10-0.37 THB/kWh

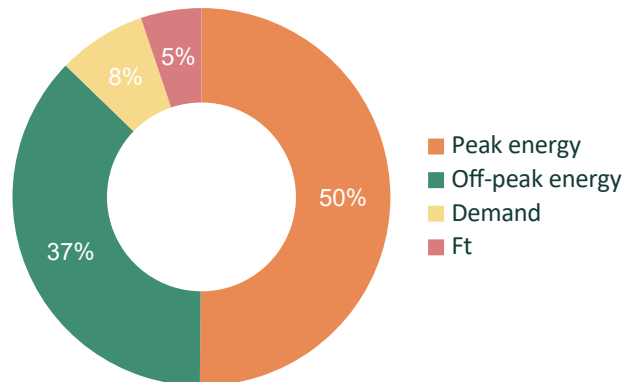
all kWh; reset every four months

Service and VAT

312.24 THB/month

then 7 percent VAT on the subtotal

June 2025 invoice: THB 2.58 million



Pre-VAT subtotal THB 2,407,767

631,580 kWh billed in the month

1,368 kW on-peak maximum demand, the single interval that sets THB 181,848 of demand charge

Ft 0.1972 THB/kWh on every kWh in that period

Energy is 87 percent of the bill; demand 8 percent, Ft 5 percent; the service charge is negligible

Why the structure matters

- Solar earns the peak rate on weekdays and the off-peak rate on weekends and holidays; the blended value of a solar kWh is well below the peak rate.
- One monthly interval sets the demand charge, so a cloudy quarter hour can wipe out a month of peak shaving unless a battery covers it.
- Ft moves every four months and is outside the site's control; the model carries 3 percent a year tariff escalation and does not forecast Ft.

Assumptions (1): technical and cost inputs

Item	Value	Basis
Usable roof share	65 percent of the outlined roof area	memo assumption kept; skylights, edges, walkways and equipment take the rest
Installed power density	0.20 kWp per usable m2	memo assumption kept; equivalent to 0.13 kWp per outlined m2
PV installed cost	500 USD/kWp	Keen quotation; below published Thai benchmarks (see slide 15)
Battery installed cost	100 USD/kW + 150 USD/kWh	Keen quotation; 325 USD/kW installed for a 1.5 hour system
Battery minimum duration	1.5 hours	provisional
PV operating cost	8.00 USD/kWp per year	supplied at 7.50 (1.5 percent of capex); optimiser rounds to whole dollars
Battery operating cost	1 percent of install cost per year	optimiser convention
Insurance	0.5 percent of capex per year	provisional
Specific yield, year 1	1,498 kWh/kWp	PVWatts irradiance for Phimai: 2,049 kWh/m2 plane of array; performance ratio 0.731; tilt 15 degrees
PV degradation	0.5 percent per year	optimiser default
Battery replacement	year 10, 100 percent of install cost, whole system	conservative case; capitalised and depreciated over 5 years
PV inverter replacement	year 11, 10 percent of PV capex	Allotrope convention, provisional
Grid export	none	strict no-export site; surplus is curtailed
Grid connection, power factor mitigation, permit fees	excluded from capex	to be added when quoted

Assumptions (2): financing, tax and scope

Financing and tax

Project life	20 years
Discount rate	11 percent, equity NPV basis
Debt	70 percent of capex, 6.5 percent, 10 years, level payments
Exchange rate	32.5 THB per USD, planning rate
Corporate income tax	20 percent flat, no incentive regime
Depreciation	5 year machinery life, PV and battery alike
Tariff escalation	3 percent a year on the PEA bill (provisional)
O&M escalation	3 percent a year
VAT	7 percent on the bill

Ownership structure

Direct ownership: the factory invests in its own roof and keeps the whole avoided PEA bill. No ESCO discount, no third-party developer margin, no new-project tax incentive. Returns are equity returns after debt service and CIT.

Emissions and scope

- Avoided emissions: avoided grid import multiplied by 0.4750 kg CO₂e per kWh (Thailand Greenhouse Gas Management Organization, 2022-2024 grid mix).
- Excluded at the client's direction: grid connection cost, power factor mitigation (no kVAR data supplied), permit and licence fees (no published schedule).
- Fifteen inputs are provisional and marked as such in the workbooks; the ones that move the answer are the usable roof share, the debt terms and the insurance rate (Appendix A).

All results are before grid connection, permitting and power-factor costs; add them when quoted.

Five configurations were optimised

1

Roof F

PV capped at 433 kWp (3,329 m²)

no storage

2

Roof F, with storage

PV capped at 433 kWp

storage sized by the optimiser

3

All roofs

PV capped at 2,122 kWp (16,319 m²)

no storage

4

All roofs, with storage

PV capped at 2,122 kWp

storage sized by the optimiser

5

No roof limit, with storage

PV free up to the 3,230 kVA connection

storage sized by the optimiser: the memo's reference optimum

Two roof groups

Roof F is one fifth of the measured area and the only roof without a structural question today. All roofs is the full site once the six older roofs pass a condition review.

How the capacity caps are built

Roof outlines on satellite
imagery

x 65% usable

x 0.20 kWp per m²

Roof F: 3,329 m² = 433
kWp

All roofs: 16,319 m² =
2,122 kWp

The memo's optimum with no roof limit is 2,847 kWp with storage; the full measured roof carries about three quarters of that, so the roof still binds. A layout drawing on the confirmed roofs will replace the 65 percent share with a panel count.

Results: the five cases side by side

Case	PV (kWp)	Storage	Capex (USD)	Year-1 savings (USD)	Grid offset	tCO2e / yr	Equity IRR	Equity NPV (USD)
1. Roof F	433	none	216,400	77,247	8.5%	304	75.5%	389,829
2. Roof F, with storage	433	33 kW / 50 kWh	227,175	79,324	8.7%	303	73.4%	394,093
3. All roofs	2,122	none	1,060,750	325,965	36.8%	1,310	62.0%	1,512,301
4. All roofs, with storage	2,122	303 kW / 634 kWh	1,186,094	356,680	39.5%	1,363	60.3%	1,612,795
5. No roof limit, with storage	2,847	507 kW / 1,881 kWh	1,756,407	460,458	50.4%	1,740	49.6%	1,855,912

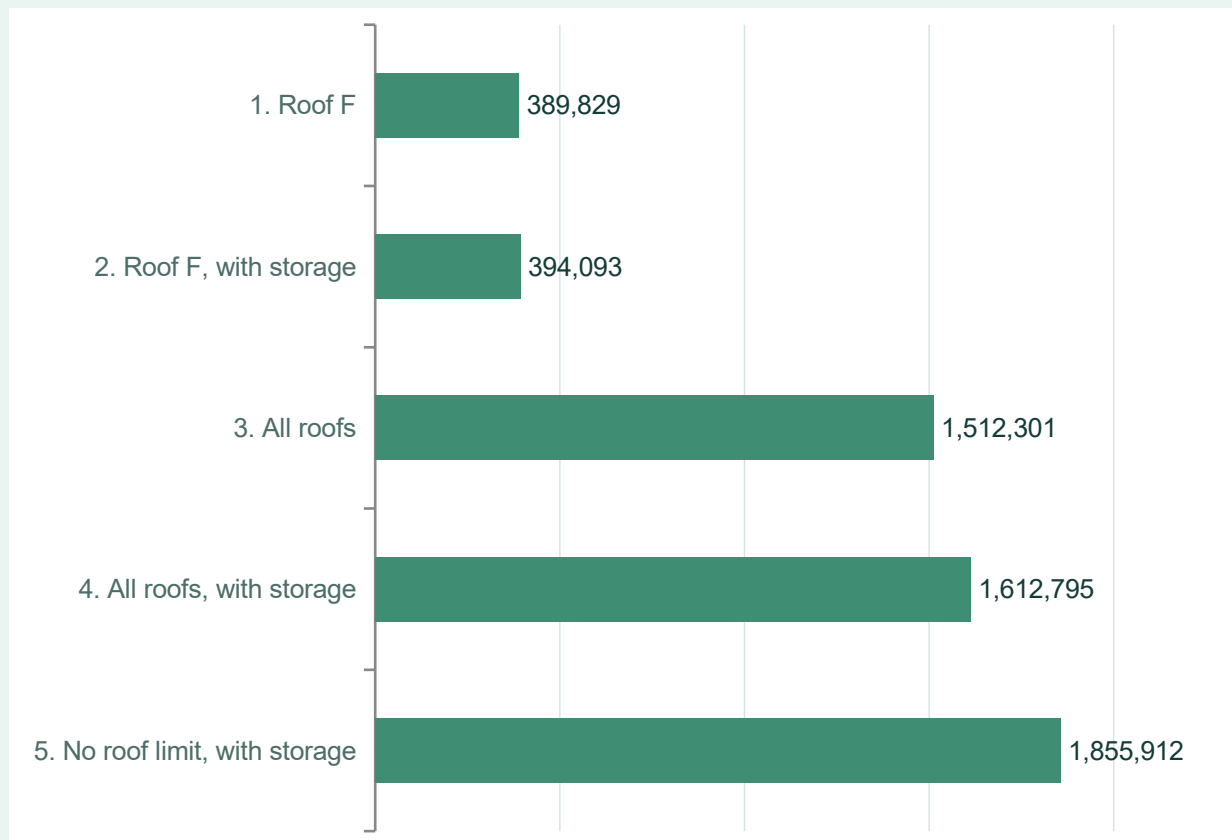
Reading the table

- Roof F alone (case 1) returns its equity in 1.4 years; its 433 kWp are almost fully used on site (1.4% curtailed).
- All roofs (case 3) multiplies capex by 4.9 and NPV by 3.9; the marginal kilowatts are used less and IRR falls from 75.5% to 62.0%.
- Storage on roof F (case 2) is a token 50 kWh worth 4,264 USD of NPV: not a project. On all roofs (case 4) it adds 125,344 USD of capex and 100,493 USD of NPV.
- Case 5, the unconstrained optimum from the memo, is shown for reference: the measured roof stops about 726 kWp short of it.

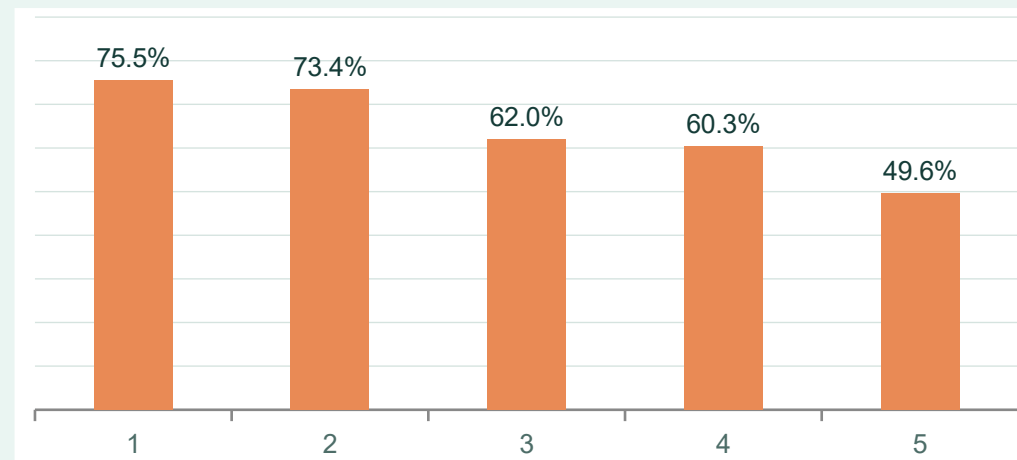
Grid offset on the memo's grid-to-load basis. The memo's cases 1 to 4 (survey-based roof widths) are retired; case 5 here is the memo's case 6, unchanged.

Bigger systems return more in total, less per dollar

Equity NPV (USD, 20 years at 11 percent)



Equity IRR



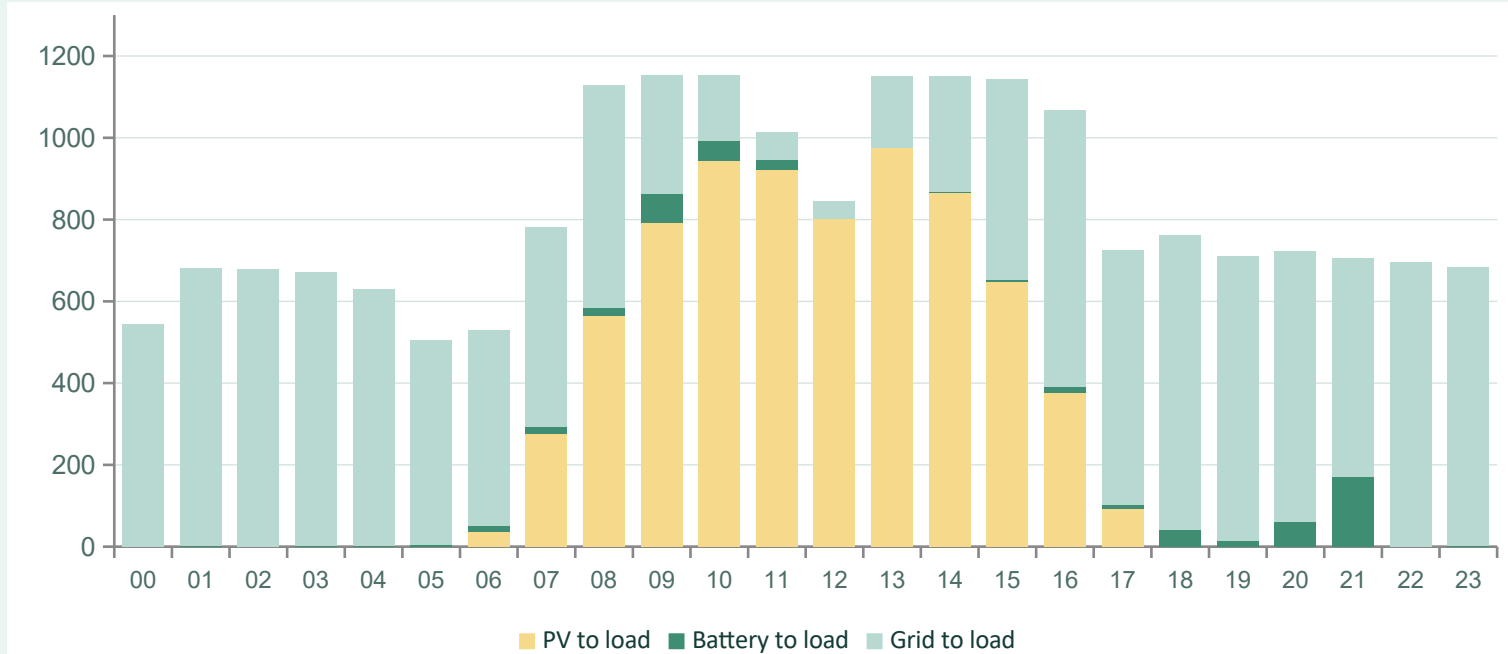
Capex and equity by case (USD)

- 1: capex 216,400, equity 64,920, debt 151,480
- 2: capex 227,175, equity 68,152, debt 159,022
- 3: capex 1,060,750, equity 318,225, debt 742,525
- 4: capex 1,186,094, equity 355,828, debt 830,266
- 5: capex 1,756,407, equity 526,922, debt 1,229,485

Simple equity payback runs from 1.4 years (case 1) to 2.0 years (case 5). These are consequences of the 500 USD/kWp input; see slide 15.

What storage buys, and why it needs the full roof

Case 4, average day: who serves the load (kW by hour)



Late morning the array also charges the battery (up to 147 kW on the average day) and still curtails up to 218 kW; the battery gives it back on the morning ramp and in the evening, inside the peak window, and tops up from the grid overnight at the off-peak rate.

The storage conclusion from the memo stands, with a size condition: the battery earns from curtailed energy and the monthly peak, and there is little of either until the array is large against the load.

193 MWh / yr

delivered by the case 4 battery to the load: about 56 percent in the evening peak hours (17:00 to 22:00), 37 percent on the morning ramp

13.2% to 9.0%

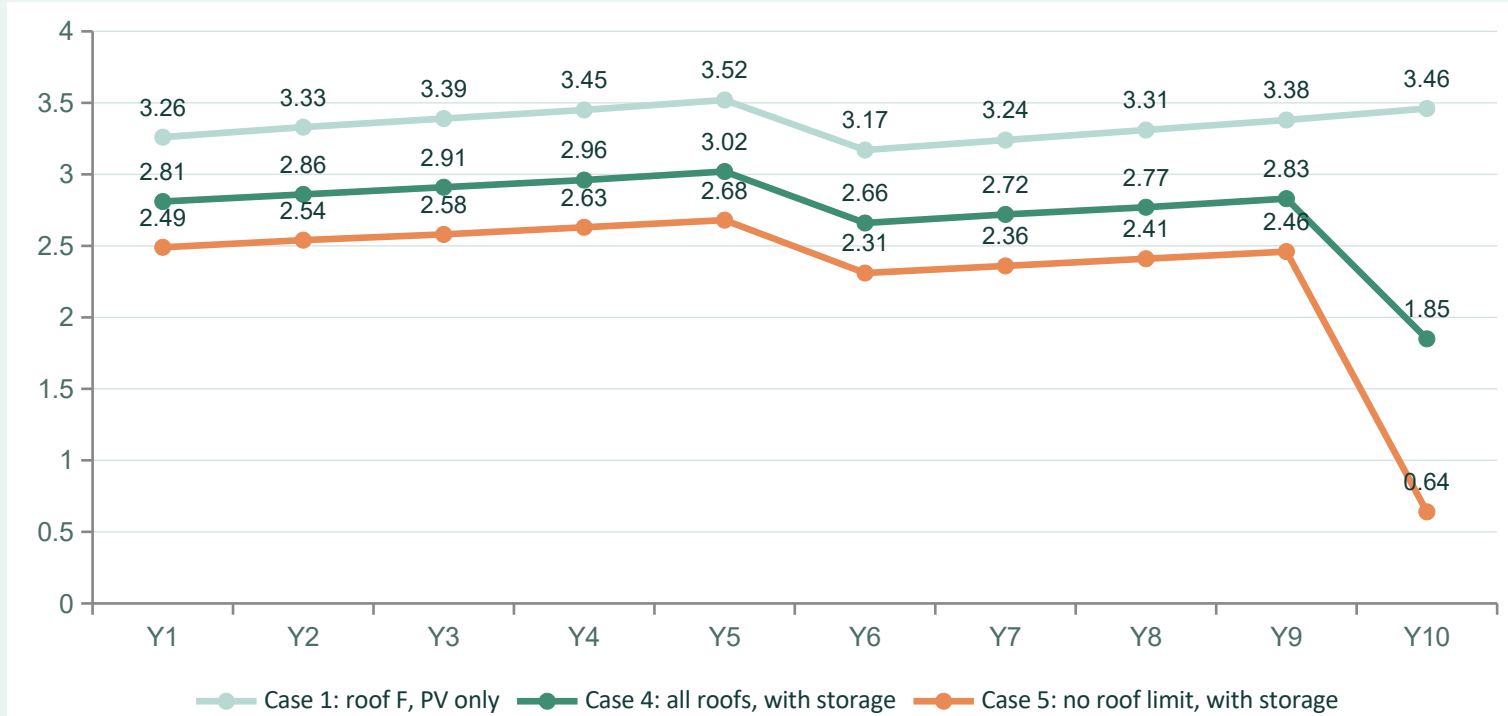
curtailment on all roofs without and with storage (cases 3 and 4): energy that would otherwise be lost

50 kWh on roof F

all the optimiser adds to the roof F array (case 2), for 4,264 USD of NPV: storage needs the full roof to earn

The year-10 battery replacement meets the final loan year

Debt service coverage ratio, years 1 to 10



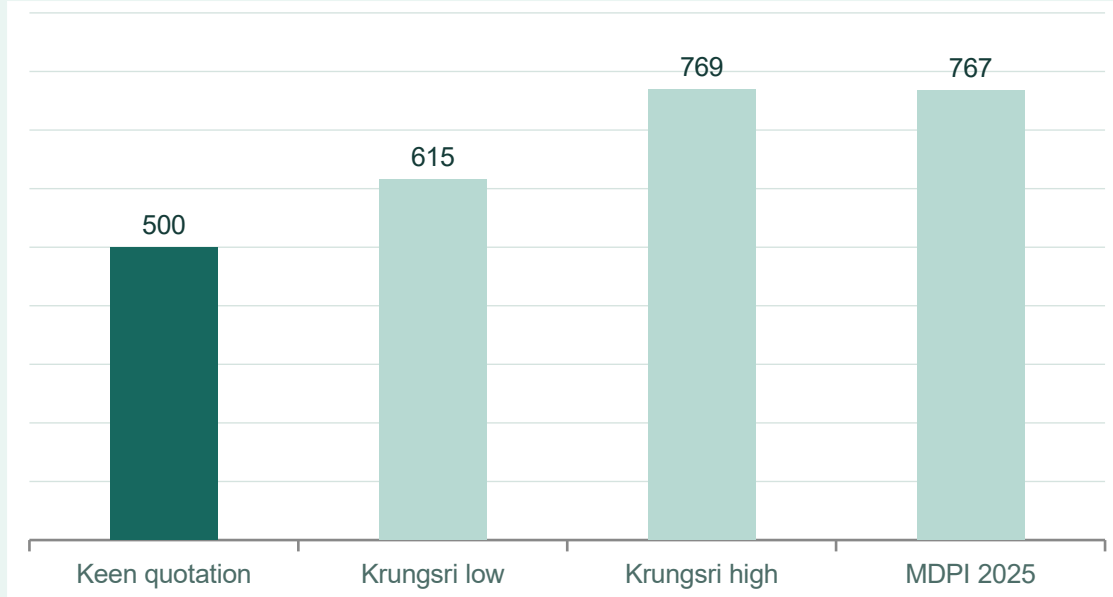
Average coverage over the loan: case 1 3.35, case 4 2.74, case 5 2.31. PV inverter replacement follows in year 11, after the loan.

A structuring point

- Case 4, year 10: cash available for debt service 213,566 USD against 115,494 USD of debt service after a 125,344 USD battery replacement; coverage 1.85 in that year.
- Still above a typical 1.2 covenant; the memo's unconstrained case (5) with a 1,881 kWh battery is the one that falls to 0.64. A reserve built in years 1 to 9, or a tenor set so the replacement falls after the final payment, removes the dip in either case.
- Roof F alone has no such year: PV only, coverage above 3.1 throughout; the inverter replacement (10 percent of PV capex) falls in year 11, after the loan.
- The whole-system replacement at 100 percent of install cost in year 10 is the memo's conservative case; a partial or cheaper replacement shrinks the dip.

Two inputs move the answer: capital cost and export

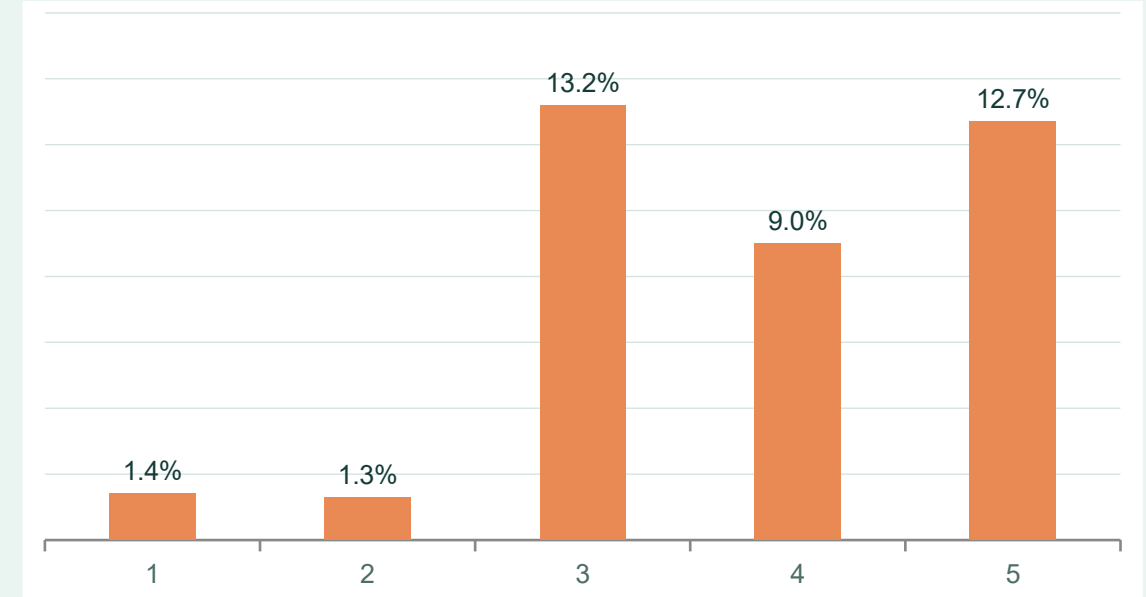
Installed PV cost, USD per kWp



The 500 USD/kWp price is used as given. It is why simple equity payback is 1.4 years on roof F; a quotation at 650 USD/kWp would change the picture materially. Confirming the price, on roof F first, is worth more than refining anything else in the analysis.

Cases: 1 roof F, 2 roof F with storage, 3 all roofs, 4 all roofs with storage, 5 no roof limit with storage.

Share of production curtailed (no export)



Roof F alone wastes almost nothing because the array is small against the daytime load. Curtailment appears with the full roof and is where an export price, if PEA or a third party offers one, would raise both the value of the larger systems and the optimal size; the question is left open rather than priced by assumption.

Recommendation: roof F now, the other roofs after review

Phase 1: roof F, PV only (case 1)

433 kWp on the roof replaced 1 to 2 years ago; no storage

Capex 216,400 USD, equity 64,920 USD

Year-1 savings 77,247 USD; grid offset 8.5%; 304 tCO₂e a year avoided

Equity NPV 389,829 USD; IRR 75.5%; payback 1.4 years; coverage above 3.1 in every year

Ready for an EPC quotation now; no structural question on this roof.

Phase 2: the other six roofs, with storage

Case 4 less case 1: 1,689 kWp more PV, 303 kW / 634 kWh of storage

Capex 969,694 USD more (case 4 total 1,186,094 USD)

Year-1 savings 279,433 USD more; site total grid offset 39.5%; 1,363 tCO₂e a year

Equity NPV 1,222,965 USD more (case 4 total 1,612,795 USD); year-10 battery replacement dips coverage to 1.85

Conditional on the older roofs passing a condition review.

Next steps, in order of value

1 Roof condition review

Structural capacity and sheeting condition of A to E and G. Four fifths of the site's solar value sits on them.

2 Quote roof F

An EPC quotation for the phase 1 array, checked against the 500 USD/kWp used here. Every return in this deck scales off that figure.

3 Decide on grid export

A price for surplus generation would raise the value of phase 2 and the optimal size; none is assumed today.

4 Supply kVAR maxima

Monthly kVAR maxima are needed to assess power factor; mitigation is currently outside the capital cost.

Then: ERC generation licence and Aor.6 building modification permit (no published fee schedule); no environmental assessment below 5 MWp. Phase 2 figures are case 4 less case 1, not a modelled sequence.

Appendix A: provisional inputs

Input	Value used	Effect if changed
Usable roof share and PV cap	65 percent of 16,319 m2 measured; 2,122 kWp all roofs, 433 kWp roof F	Sets the size of every roof case; a layout drawing will replace the share
Roof condition, A to E and G	assumed able to carry panels once reviewed	A failed review removes those roofs from phase 2 or adds re-roofing cost
Debt fraction	70 percent of capex	Moves equity IRR and DSCR; NPV to equity changes with leverage
Debt term	10 years	A longer tenor takes the year-10 replacement past the final payment
Insurance rate	0.5 percent of capex per year	Recurring cost; second-order at this scale
PEA tariff escalation	3 percent per year	Raises or lowers every year's savings; Ft is not forecast
Battery minimum duration	1.5 hours	Shapes the kW to kWh ratio the optimiser may choose
Battery O&M	1 percent of install cost per year	Small; offset by the conservative replacement assumption
PV inverter replacement	year 11, 10 percent of PV capex	Allotrope convention; a different year or share moves one year's cash flow
PV tilt	15 degrees	Pitch not recorded; yield sensitivity is small at this latitude
Grid connection cost	excluded; placeholder pending quotation	Add when quoted
Permitting and licence fees	excluded (no published schedule)	Add when quoted
Power factor mitigation	excluded (no kVAR data)	Conditional charge of 56.07 THB per kVAR above the allowance

Counted as fifteen items on the workbook because the roof area and the PV cap are listed separately, as are the PV inverter year and share; the roof condition line is added here.

Optimisation

- REopt (NREL) v0.57 through Allotrope's REopt API, HiGHS solver.
- 15 minute resolution, one metered year of load (2025).
- PVWatts irradiance for Phimai; tilt 15 degrees; 0.5 percent a year PV degradation.
- Objective: lowest 20 year cost of electricity for the site; no export; storage may charge from grid or PV.

Pro forma

- 20 year cash flow, direct ownership, 70 percent debt at 6.5 percent over 10 years.
- Thai CIT 20 percent flat; 5 year machinery depreciation; 7 percent VAT on the bill.
- Battery replaced in year 10 at 100 percent of install cost; PV inverter in year 11 at 10 percent of PV capex; both capitalised.
- Every workbook reconciles its live formulas to the engine (audit sheet).

Emissions and permitting

- Avoided import x 0.4750 kg CO₂e per kWh (TGO 2022-2024 grid mix, effective 1 January 2026).
- The optimiser's emissions outputs use US datasets and are not used.
- ERC generation licence and Aor.6 building modification permit required; no published fee schedules.
- IEE at 5 MWp, EIA at 10 MWp: not triggered below 3 MWp.

Files: five pro forma workbooks (one per case) under rofu_thailand_v2, the satellite roof measurement, the memo of 12 September 2026 (survey-based roof; superseded on the roof only) and its Vietnamese translation, the tariff structure briefing, and the case inputs, solver payloads and raw results for each case.